

Comparison of guidelines and consensus articles on the management of patients with acne with oral isotretinoin

Abstract

Guidelines and consensus statements on the management of acne aim to provide evidence-based, expert-driven recommendations. This review compares current guidelines and consensus articles to compile recommendations on the use of oral isotretinoin in acne treatment. Ten common and clinically relevant questions are addressed, including indications for isotretinoin, recommended daily and cumulative doses, treatment duration, and laboratory monitoring. Special considerations such as timing of dermatologic procedures and potential associations with depression or inflammatory bowel disease (IBD) are also evaluated. A major limitation across guidelines is the use of differing acne classification systems. The recommended daily dose ranges from 0.3 to 0.5 mg/kg in European guidelines to up to 1 mg/kg in U.S. guidelines. Only European guidelines specify a minimum treatment duration of 6 months. All guidelines emphasize strict pregnancy prevention due to teratogenicity. Monitoring for symptoms of depression is recommended in European, French, and U.S. guidelines. Key clinical issues that remain inconsistently addressed include age indications, target cumulative dose, timing of procedures, isotretinoin-IBD association, acne flare prevention, and laboratory monitoring protocols. These topics require clearer inclusion in future guidelines, as they frequently arise in clinical practice.

Introduction

Oral isotretinoin (13-cis retinoic acid), a natural metabolite of vitamin A, was approved for the treatment of acne vulgaris in 1982 by the U.S. Food and Drug Administration (FDA) and in 1983 by the European Medicines Agency (EMA). Isotretinoin undergoes rapid intracellular conversion to all-trans retinoic acid and exhibits strong biological activity with low binding affinity for retinoid receptors. It effectively inhibits sebocyte proliferation, differentiation, and lipid synthesis both in vivo and in vitro, reduces *Cutibacterium acnes* (formerly *Propionibacterium acnes*) colonization, and decreases inflammation and hyperkeratinization within the pilosebaceous unit.

The efficacy and safety of oral isotretinoin were evaluated in a systematic review by Vallerand et al., which included 11 randomized controlled trials published up to 2016. All studies demonstrated a clinically significant reduction in acne lesions with isotretinoin compared to control groups (placebo, systemic antibiotics, or other comparators). Adverse events occurred approximately twice as often with isotretinoin versus controls. Most were cutaneous and related to xerosis. Serious adverse events leading to discontinuation included Stevens-Johnson syndrome, severe cheilitis, photophobia, elevated liver enzymes, anorexia, headache, and depression.

This review compares current international and national guidelines and consensus statements on the use of oral isotretinoin for acne. Guidelines are intended to optimize patient care through evidence-based, critically appraised recommendations on treatment benefits and risks. Key areas of focus include indications for use, dosing strategies (daily and cumulative), treatment duration, laboratory monitoring, and special considerations such as procedural timing and potential psychiatric or gastrointestinal risks.

Methods

Eligible publications included updated international and national guidelines and consensus articles containing recommendations on oral isotretinoin for acne. EMA recommendations were included as they inform drug prescribing frameworks in Europe. Guidelines published before 2012 or without a stated

validity date were excluded. Only publications in English or French were considered.

Search terms included: *isotretinoin*, *systemic treatment*, *acne*, *guidelines*, *consensus*. Reference lists of included articles were manually screened for additional relevant guidelines or consensus documents. The final literature search was conducted on 20 May 2019.